

Real-Time Face Detection System

Step-by-step guide to install and run the project on Windows (PowerShell / Command Prompt)

Step 1: Install Python

Download Python 3.12 (or any version from 3.9 to 3.13) from python.org. During installation, tick **Add Python to PATH**. Then check the installation:

```
python --version
```

Step 2: Open the project folder in the terminal

Extract **face_detection_project.zip**. Open the folder in File Explorer, click the address bar, type **cmd** and press Enter. Or use the commands below (change the path to your folder):

```
cd C:\Users\YourName\Desktop\face_detection_project
dir
```

The **dir** command must show **main.py**. If not, you are in the wrong folder.

Step 3: Install OpenCV (version 4.x)

The project needs OpenCV 4.x. If a newer 5.x version is already installed, remove it first, then install the correct one:

```
pip uninstall -y opencv-python
pip install "opencv-python<5"
```

Or install from the file: **pip install -r requirements.txt**

Step 4: Check the installation

This should print a version number starting with 4 and no error:

```
python -c "import cv2; print(cv2.__version__)"
```

Step 5: Run the project

Start the program. If Windows asks for camera permission, click **Allow**.

```
python main.py
```

Step 6: Test face detection

Look at the camera. A green box appears around your face with the text **Face Detected** and **Total Faces: 1**. Bring a second person into view and the count increases. With no face, the text shows **No Face Detected**.

Step 7: Stop the program

Press **q** or **ESC** on the keyboard, or click the red **STOP** button in the video window. The webcam is released safely.

Troubleshooting

Problem / error	Solution
'pip' is not recognised	<code>python -m pip install "opencv-python<5"</code>
module 'cv2' has no attribute 'CascadeClassifier'	OpenCV 5.x is installed. Repeat Step 3.
Cannot open the webcam	Close Zoom/Teams/Camera app, allow camera access in Windows Settings > Privacy > Camera, or change CAMERA_INDEX to 1 in main.py.
Could not open requirements.txt	You are in the wrong folder. Use <code>dir</code> and <code>cd</code> to reach the project folder, or use <code>pip install "opencv-python<5"</code> .
Still not working	Install Python 3.12, then repeat Steps 3 to 5.